

# EDA Toolkit: A Unified Framework for Transparent and Reproducible Data Science Workflows



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# | AI Hits 94%. The Patient Still Loses.

## WHAT THE FIELD REPORTS

93–94%

Reported classification accuracy is routine in clinical machine learning papers.

## What that number doesn't tell you

01

**High accuracy is not the same as clinical signal.**

A model can reach 94% on a dataset with no underlying biological structure.

02

**Most clinical ML papers skip data validation.**

Feature selection and modeling get the attention. Distributional checks rarely happen first.

03

**Bad data still hits the metric.**

It just stops being useful the moment it leaves the paper.

*Trust starts before the model. With the data.*

# EDA Toolkit: Validation Before Modeling

Two checks that tell you, within minutes, whether your dataset actually contains the signal you are claiming to predict. You will see both in action in the next two slides.

## Class Separability

Plots feature distributions by outcome group. If a feature distinguishes the two groups, you see two curves. If not, one overlapping mass.

*No statistics required. Your eyes do the work.*

## Cohort Tables

The publication-ready Table 1 every clinical paper expects, with statistical testing built in. Surfaces demographic patterns that signal a synthetic dataset.

*The view a reviewer or regulator would demand.*

*Two checks. One toolkit. Built so this step never gets skipped again.*



# Case Study: 322,895 Records. 94% Accuracy.

A 2025 IEEE paper applied ensemble feature selection to pediatric Type 1 Diabetes data, reporting 93.58 to 94.13% classification accuracy.

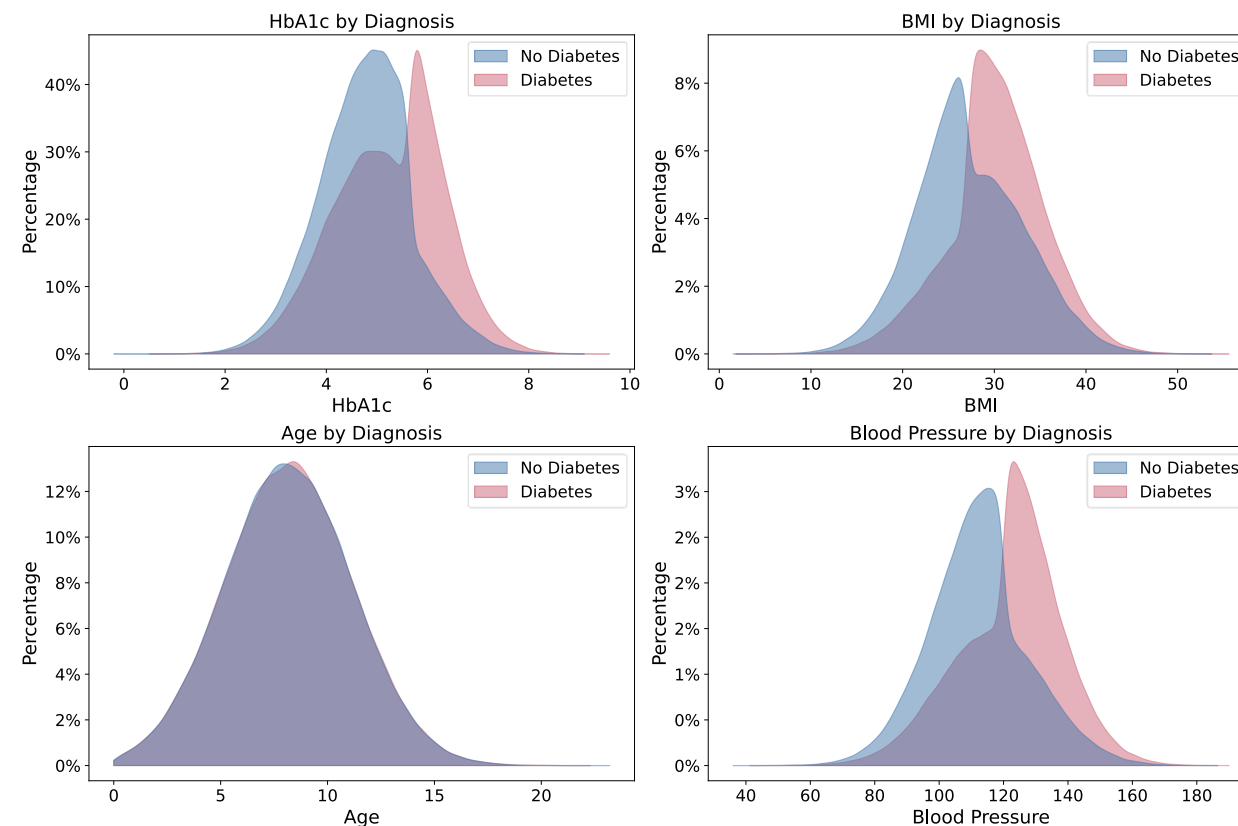
## WHAT EDA TOOLKIT FOUND

- HbA1c, the primary T1D biomarker, shows no class separability.
- Top-ranked features (Pancreatic, BMI) have identical distributions across diagnostic groups.
- All feature pairwise correlations approximate zero. Labels appear independent of values.

## VERDICT

**The accuracy is real. The signal is not.**

*Synthetic data artifacts, not clinical relationships.*



Probability density distributions of HbA1c, BMI, Age, Blood Pressure by diagnostic class. Near-perfect overlap across all four features.

# Table 1: The Statistical Tell

Same dataset. Different lens. `generate_table1()` reveals what the headline accuracy hides.

| Variable            | Overall          | Diabetes<br>(n = 133,768) | No Diabetes<br>(n = 189,127) | P-value |
|---------------------|------------------|---------------------------|------------------------------|---------|
| BMI                 | 28.50 (5.87)     | 30.24 (5.37)              | 27.27 (5.90)                 | < 0.001 |
| HbA1c               | 4.97 (1.02)      | 5.26 (1.07)               | 4.77 (0.93)                  | < 0.001 |
| Blood pressure      | 116.41 (16.67)   | 122.45 (16.21)            | 112.14 (15.64)               | < 0.001 |
| Pancreatic          | 0.97 (0.73)      | 0.97 (0.73)               | 0.97 (0.74)                  | 0.36    |
| Gender              | 322,895 (100%)   | 133,768 (41.43%)          | 189,127 (58.57%)             | 0.07    |
| Male                | 161,577 (50.04%) | 67,189 (50.23%)           | 94,388 (49.91%)              |         |
| Female              | 161,318 (49.96%) | 66,579 (49.77%)           | 94,739 (50.09%)              |         |
| Age group           | 322,895 (100%)   | 133,768 (41.43%)          | 189,127 (58.57%)             | 0.43    |
| Infant (0-1)        | 5,421 (1.68%)    | 2,233 (1.67%)             | 3,188 (1.69%)                |         |
| Toddler (2-4)       | 40,173 (12.44%)  | 16,733 (12.51%)           | 23,440 (12.39%)              |         |
| Child (5-11)        | 243,485 (75.41%) | 100,697 (75.28%)          | 142,788 (75.50%)             |         |
| Adolescent (12-17)  | 33,622 (10.41%)  | 14,014 (10.48%)           | 19,608 (10.37%)              |         |
| Young adult (18-20) | 189 (0.06%)      | 89 (0.07%)                | 100 (0.05%)                  |         |
| Adult (21+)         | 5 (0.00%)        | 2 (0.00%)                 | 3 (0.00%)                    |         |

## Red < 0.001

Welch's t-test flags BMI, HbA1c, and Blood Pressure as highly significant. Those p-values are real numerical output. The signal is not.

## Pancreatic: $p = 0.36$

The paper's top-ranked predictive feature shows no statistical separation between groups. A null result at the very top of the rankings.

## Identical group proportions

50.04% Male in both diabetic and non-diabetic groups. 75.4% Child (5–11) in both. Composition is identical across outcome classes. That is the synthetic data fingerprint.

# | Plug It Into Your Pipeline

EDA Toolkit is open source and works with any tabular clinical dataset. Here is who should be using it tomorrow.

## | Pharma & Biotech

Validate dataset integrity before committing pipeline budget. Catch synthetic, corrupted, or label-noisy data before it propagates into go/no-go decisions on candidates.

## | Clinical Researchers

Don't accept high accuracy as evidence of clinical utility. Demand distributional validation before any published model influences patient care.

## | Investors

Treat upstream EDA as due diligence. The same way you audit financials, audit the data backing your portfolio companies' ML claims.

**`pip install eda-toolkit` • [pypi.org/project/eda-toolkit](https://pypi.org/project/eda-toolkit) • [datasciencedynamics.com](https://datasciencedynamics.com)**

*Best Poster Award • CHOC Pediatric & Lifespan Data Science Conference 2026*

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# References

*The paper analyzed in the case study, and the toolkit used for the analysis.*

1

## PAPER ANALYZED IN THE CASE STUDY

Syahidin Y, Maulidevi NU, Surendro K. Ensemble feature selection: a hybrid approach for pediatric type 1 diabetes risk factors. 2025 International Conference on Smart-Green Technology in Electrical and Information Systems (ICSGTEIS); 2025 Oct 9-10; Bali, Indonesia. IEEE; 2025. p. 30-5.

DOI: [10.1109/ICSGTEIS68532.2025.11284421](https://doi.org/10.1109/ICSGTEIS68532.2025.11284421)

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## TOOLKIT USED FOR THE ANALYSIS

Shpaner L, Gil O. EDA Toolkit [software]. Version 0.0.29. Zenodo; 2024 Aug.

DOI: [10.5281/zenodo.13162633](https://doi.org/10.5281/zenodo.13162633)

*Full citation list available on the poster at [datasciencedynamics.com](https://datasciencedynamics.com).*

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# Thank you.

*Questions, conversations, collaborations welcome.*

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